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Medicament delivery device

Abstract

A medicament delivery device comprises: a needle for injecting a medicament; a body; a needle cover axially movable relative to the body between an extended position and a retracted position; a needle cover biasing member configured to bias the needle cover distally; a medicament delivery mechanism; and a latch movable by the needle cover between an engaged configuration, in which the latch prevents delivery of the medicament, and a disengaged configuration, in which the latch allows delivery of the medicament, wherein the needle cover is configured such that: proximal movement of the needle cover from the extended position towards the retracted position moves the latch from the engaged configuration to the disengaged configuration; and distal movement of the needle cover from the retracted position towards the extended position moves the latch from the disengaged configuration to the engaged configuration.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4989307	12/1990	Sharpe et al.	N/A	N/A
5069667	12/1990	Freundlich et al.	N/A	N/A
5092462	12/1991	Sagstetter et al.	N/A	N/A
5356385	12/1993	Latini	N/A	N/A
6224567	12/2000	Roser	N/A	N/A
8376993	12/2012	Cox et al.	N/A	N/A
2013/0096513	12/2012	Smith	N/A	N/A
2015/0265776	12/2014	Beek et al.	N/A	N/A
2016/0129188	12/2015	Kiilerich	N/A	N/A
2017/0246396	12/2016	Wei	N/A	N/A
2020/0001014	12/2019	Holmqvist	N/A	N/A
2021/0162137	12/2020	Hagihira et al.	N/A	N/A
2021/0236714	12/2020	Limaye	N/A	N/A
2022/0288299	12/2021	Limaye et al.	N/A	N/A
2023/0001101	12/2022	Larsen	N/A	A61M 5/3202
2023/0277769	12/2022	Chang	N/A	N/A
2025/0018119	12/2024	Lin et al.	N/A	N/A
2025/0041529	12/2024	Kwolek et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2024089	12/1969	FR	N/A
WO 2014/060369	12/2013	WO	N/A
WO 2014/198858	12/2013	WO	N/A
WO 2018/146589	12/2017	WO	N/A
WO 2018/215271	12/2017	WO	N/A
WO 2021/122192	12/2020	WO	N/A
WO 2021/122196	12/2020	WO	N/A
WO 2022/023011	12/2021	WO	N/A
WO 2022/117671	12/2021	WO	N/A
WO 2022/117683	12/2021	WO	N/A
WO 2022/175241	12/2021	WO	N/A
WO 2022/175242	12/2021	WO	N/A
WO 2022/175245	12/2021	WO	N/A
WO 2022/175246	12/2021	WO	N/A
WO 2022/175247	12/2021	WO	N/A

OTHER PUBLICATIONS

Needle-based injection systems for medical use requirements and test methods, Part 1: Needle injection systems, ISO 11608 1:2014(E), Third Edition, Switzerland, ISO, Dec. 15, 2014, pp. 1-13. cited by applicant
U.S. Appl. No. 18/819,497, filed Aug. 29, 2024, Timothy Denyer. cited by applicant

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Background/Summary

TECHNICAL FIELD
(1) This application relates to a medicament delivery device for delivery of a medicament, and a method of using a medicament delivery device.
BACKGROUND
(2) Drug delivery devices such as auto-injectors are used to deliver a range of medicaments. In an auto-injector device, some or all of the actions required to use the injector device in administering medicament are automated.
(3) An auto-injector device may have a needle cover which is axially movable to cover and uncover a needle, with the needle cover being biased by a spring to extend over the needle. Typically, the user presses the needle cover against an injection site, against the force of the spring, to push the needle cover into the housing and to uncover the needle which is pushed into the injection site. Medicament is automatically dispensed from the needle via an automated mechanism. A user must typically hold the needle cover in a holding position for a predetermined period of time, to ensure that the correct dose of medicament is dispensed from the device, before removing the device from the injection site.
(4) Some users may experience discomfort during the medicament delivery process. For example, some users may experience discomfort in the vicinity of the injection site when the delivered dose of medicament is large and/or delivered over a long period of time. If the discomfort becomes excessive, these users may decide to remove the auto-injector device from the injection site prematurely, which may result in incomplete delivery of the medicament dose, additional discomfort from premature withdrawal of the needle, and/or a wet injection site from leaked medicament.
(5) The present disclosure provides an injector device that addresses one or more of the problems mentioned above, and to provide an improved injector device.
SUMMARY
(6) A first aspect of this disclosure provides medicament delivery device comprising: a needle for injecting a medicament; a body having a proximal end and a distal end; a needle cover axially movable relative to the body between an extended position, in which a distal end of the needle cover is distal to a distal end of the needle, and a retracted position, in which the distal end of the needle is distal to the distal end of the needle cover; a needle cover biasing member configured to bias the needle cover distally; a medicament delivery mechanism; and a latch movable by the needle cover between an engaged configuration, in which the latch is engaged with a component of a medicament delivery mechanism to prevent delivery of the medicament, and a disengaged configuration, in which the latch is disengaged from the component of the medicament delivery mechanism to allow delivery of the medicament, wherein the needle cover is configured such that: proximal movement of the needle cover from the extended position towards the retracted position

moves the latch from the engaged configuration to the disengaged configuration; and distal movement of the needle cover from the retracted position towards the extended position moves the latch from the disengaged configuration to the engaged configuration.
(7) The latch may be movable between the engaged position and the disengaged position a plurality of times for pausing and resuming delivery of the medicament a plurality of times.
(8) The medicament delivery device may further comprise a needle cover guide comprising a track, wherein the needle cover comprises an arm configured to engage the track such that a proximal movement of the needle cover is limited after the needle cover has moved from the retracted position to the extended position.
(9) The needle cover guide may be rotatable relative to the body, wherein the track extends at least partially around a circumference of the needle cover guide for rotating the needle cover guide as the needle cover moves from the extended position to the retracted position.
(10) The flexible arm may comprise a protrusion configured to engage the track.
(11) The track may comprise a first region and a second region, wherein the protrusion travels from the first region to the second region as the needle cover moves from the extended position to the retracted position. The first region may extend at least partially around the circumference of the needle cover guide.
(12) The track may further comprise a third region, the protrusion may travel from the second region to the third region as the needle cover moves distally from the retracted position to the extended position, and the third region may comprise a locking element configured to limit proximal movement of the needle cover when the needle cover is in the extended position.
(13) The locking element may comprise a locking surface configured to be engaged by the needle cover when the needle cover is in the extended position to limit proximal movement of the needle cover.
(14) The track may further comprise a fourth region, wherein the protrusion travels from the third region to the fourth region as the needle cover guide is rotated, wherein proximal movement of the needle cover from the locked position to the retracted position is allowed when the protrusion is in the fourth region.
(15) The medicament delivery device may further comprise an actuation member, wherein the actuation member is actuatable by a user to rotate the needle cover guide relative to the needle cover such that the protrusion travels from the third region to the fourth region.
(16) At least one of the actuation member or the needle cover guide may comprise a ramped surface configured to convert linear movement of the actuation member into rotation of the needle cover guide.
(17) The actuation member may comprise a button arranged at a proximal end of the medicament delivery device.
(18) The latch may comprise an engaging element configured to: engage an engaging element of the component of the medicament delivery mechanism when the latch is in the engaged position; and be disengaged from the engaging element of the component when the latch is in the disengaged position.
(19) The engaging element of the latch may comprise a projection and the engaging element of the component may comprise a recess.
(20) The component of the medicament delivery mechanism may be a collar that is rotatable relative to the body to dispense the medicament, wherein the latch is configured to engage the collar when in the engaged position to limit rotation of the collar.
(21) The latch may be biased from the engaged configuration to the disengaged configuration, wherein the latch is held in the engaged position by the needle cover when the needle cover is in the extended position, and wherein the needle cover comprises a recess configured to receive at least a portion of the latch when the needle cover is in the retracted position such that the latch can move to the disengaged position.

(22) The medicament delivery device may further comprise a needle cover lock that is movable by the medicament delivery mechanism between an initial configuration, in which movement of the needle cover from the extended position to the retracted position is not limited by the needle cover lock, and a locking position, in which movement of the needle cover from the extended position to the retracted position is limited by the needle cover lock.
(23) The needle cover lock may be moved from the disengaged position to the engaged position by a plunger of the medicament delivery mechanism.
(24) The medicament delivery device may further comprise the medicament. The medicament may be held in a container of the medicament delivery device.
(25) A second aspect of this disclosure provides a method of operating a medicament delivery device disclosed herein, the method comprising: moving a needle cover of the medicament delivery device from an extended position to a retracted position to initiate delivery of a medicament; moving the needle cover of the medicament delivery device from the retracted position to the extended position to pause the delivery of the medicament; and resuming the delivery of the medicament by: actuating an actuation member of the medicament delivery device to rotate a needle cover guide; and subsequent to actuating the actuation member, moving the needle cover from the extended position to the retracted position.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) Exemplary embodiments of the present disclosure are described with reference to the accompanying drawings, in which:
(2) FIG. **1**A shows an injector device with a cap attached;
(3) FIG. **1**B shows the injector device of FIG. **1**A with the cap removed;
(4) FIG. **2** is a cross-sectional view of a medicament delivery device;
(5) FIG. **3**A is a schematic cross-sectional view of a medicament delivery device in accordance with one or more embodiments, in an initial state;
(6) FIG. **3**B is a schematic cross-sectional view of the medicament delivery device of FIG. **3**A, in an holding state;
(7) FIG. **3**C is a schematic cross-sectional view of the medicament delivery device of FIG. **3**B, in a paused state;
(8) FIG. **3**D is a schematic cross-sectional view of the medicament delivery device of FIG. **3**C, in a resetting state;
(9) FIG. **3**E is a schematic cross-sectional view of the medicament delivery device of FIG. **3**D, in an resumed state;
(10) FIG. **3**F is a schematic cross-sectional view of the medicament delivery device of FIG. **3**E, in a locked state;
(11) FIG. **4**A is a schematic cross-sectional view of a portion of the medicament delivery device of FIG. **3**A, showing a latch in an engaged configuration with a collar;
(12) FIG. **4**B is a schematic cross-sectional view of a portion of the medicament delivery device of FIG. **3**B, showing the latch in a disengaged configuration with the collar;
(13) FIG. **4**C is a schematic cross-sectional view of a portion of the medicament delivery device of FIG. **3**C, showing the latch in the engaged configuration with the collar;
(14) FIG. **5**A is a partial perspective view of the needle cover guide, needle cover and actuation member of the medicament delivery device of FIG. **3**A;
(15) FIG. **5**B is a partial perspective view of the needle cover guide, needle cover and actuation member of the medicament delivery device of FIG. **3**B;
(16) FIG. **5**C is a partial perspective view of the needle cover guide, needle cover and actuation

member of the medicament delivery device of FIG. **3**C;

(17) FIG. **5**D is a partial perspective view of the needle cover guide, needle cover and actuation member of the medicament delivery device of FIG. **3**D, during actuation of the actuation member;

(18) FIG. **5**E is a partial perspective view of the needle cover guide, needle cover and actuation member of the medicament delivery device of FIG. **5**D, after release of the actuation member;

(19) FIG. **6**A is a schematic partial side view of the medicament delivery device of FIG. **3**A, showing a needle cover lock in an initial configuration and the needle cover in an extended position;

(20) FIG. **6**B is a schematic partial side view of the medicament delivery device of FIG. **3**B, showing the needle cover lock in the initial configuration and the needle cover in a retracted position;

(21) FIG. **6**C is a schematic partial side view of the medicament delivery device of FIG. **3**E, showing the needle cover lock in a locking configuration and the needle cover in the retracted position;

(22) FIG. **6**D is a schematic partial side view of the medicament delivery device of FIG. **3**F, showing the needle cover lock in the locking configuration and the needle cover in the extended position; and

(23) FIG. **7** is a flowchart illustrating a method of using a medicament delivery device in accordance with one or more aspects.

DETAILED DESCRIPTION

(24) Aspects of the present disclosure can allow a user to initiate, pause and/or resume the delivery of a medicament by a medicament delivery device.

(25) A drug delivery device (also referred to as a medicament delivery device), as described herein, may be configured to inject a medicament into a subject (e.g., a patient). For example, delivery could be sub-cutaneous, intra-muscular, or intravenous. Such a device could be operated by the subject themselves (i.e., ‘self-administration’) or by a different user, such as a nurse or physician providing care to the subject, and can include various types of safety syringe, pen-injector, or auto-injector. The device can include a cartridge-based system that requires piercing a sealed ampule before use. Volumes of medicament delivered with these various devices can range from about 0.5 ml to about 2 ml (e.g., to about 2.25 ml). Yet another device can include a large volume device (“LVD”) or patch pump (in some instances referred to as an on body injector (OBI) or on body device (OBD)), configured to adhere to a subject's skin for a period of time (e.g., about 5, 15, 30, 60, or 120 minutes) to deliver a “large” volume of medicament (typically about 2 ml to about 10 ml, or about 2 mL to about 20 mL).

(26) In combination with a specific medicament, the presently described devices may also be customized in order to operate within required specifications. For example, the device may be customized to inject a medicament within a certain time period (e.g., about 3 to about 20 seconds for auto-injectors, and about 10 minutes to about 60 minutes for an LVD). Other specifications can include a low or minimal level of discomfort, or to certain conditions related to human factors, shelf-life, expiry, biocompatibility, environmental considerations, etc. Such variations can arise due to various factors, such as, for example, a drug ranging in viscosity from about 3 cP to about 50 cP. Consequently, a drug delivery device will often include a hollow needle ranging from about 25 to about 31 Gauge in size. Common sizes are 27 and 29 Gauge.

(27) The delivery devices described herein can also include one or more automated functions. For example, one or more of needle insertion, medicament injection, and needle retraction can be automated. Energy for one or more automation steps can be provided by one or more energy sources. Energy sources can include, for example, mechanical, pneumatic, chemical, or electrical energy. For example, mechanical energy sources can include springs, levers, elastomers, or other mechanical mechanisms to store or release energy. One or more energy sources can be combined into a single device. Devices can further include gears, valves, or other mechanisms to convert

energy into movement of one or more components of a device.
(28) The one or more automated functions of an auto-injector may each be activated via an activation mechanism. Such an activation mechanism can include one or more of a button, a lever, a needle sleeve (also referred to as a needle cover), or other activation component. Activation of an automated function may be a one-step or multi-step process. That is, a user may need to activate one or more activation components in order to cause the automated function. For example, in a one-step process, a user may depress a needle sleeve against their body in order to cause injection of a medicament. Other devices may require a multi-step activation of an automated function. For example, a user may be required to depress a button and retract a needle sleeve in order to cause injection.
(29) In addition, activation of one automated function may activate one or more subsequent automated functions, thereby forming an activation sequence. For example, activation of a first automated function may activate at least two of needle insertion, medicament injection, and needle retraction. Some devices may also require a specific sequence of steps to cause the one or more automated functions to occur. Other devices may operate with a sequence of independent steps.
(30) Some delivery devices can include one or more functions of a safety syringe, pen-injector, or auto-injector. For example, a delivery device could include a mechanical energy source configured to automatically inject a medicament (as typically found in an auto-injector) and a dose setting mechanism (as typically found in a pen-injector).
(31) According to some embodiments of the present disclosure, an exemplary drug delivery device **10** is shown in FIGS. **1**A & **1**B. Device **10**, as described above, is configured to inject a medicament **15** into a subject's body. Device **10** includes a housing **11**, which may also be referred to as a body, which typically contains a reservoir **14** containing the medicament **15** to be injected (e.g., a syringe) and the components required to facilitate one or more steps of the delivery process. Device **10** can also include a cap assembly **12** that can be detachably mounted to the housing **11**. Typically, a user must remove cap **12** from housing **11** before device **10** can be operated. Device **10** can include a window **11***a* through which a user may view medicament **15** remaining in the reservoir **14**.
(32) As shown, housing **11** is substantially cylindrical and has a substantially constant diameter along the longitudinal axis X. The housing **11** has a distal region **20** and a proximal region **21**. The term “distal” refers to a location that is relatively closer to a site of injection, and the term “proximal” refers to a location that is relatively further away from the injection site.
(33) Device **10** can also include a needle sleeve **13** coupled to housing **11** to permit movement of the needle sleeve **13** relative to housing **11**. For example, needle sleeve **13** can move in a longitudinal direction parallel to longitudinal axis X. Specifically, movement of needle sleeve **13** in a proximal direction can permit a needle **17** to extend from distal region **20** of housing **11**.
(34) Insertion of needle **17** can occur via several mechanisms. For example, needle **17** may be fixedly located relative to housing **11** and initially be located within an extended needle sleeve **13**. Proximal movement of needle sleeve **13** by placing a distal end of sleeve **13** against a subject's body and moving housing **11** in a distal direction will uncover the distal end of the needle **17**. Such relative movement allows the distal end of the needle **17** to extend into the injection site, such as a portion of the subject's body. Such insertion is termed “manual” insertion as needle **17** is manually inserted via the subject's manual movement of housing **11** relative to the needle sleeve **13**.
(35) Another form of insertion is “automated,” whereby the needle **17** moves relative to housing **11**. Such insertion can be triggered by movement of the needle sleeve **13** or by another form of activation, such as, for example, a button **22**. As shown in FIGS. **1**A & **1**B, the button **22** is located at a proximal end of housing **11**. However, in other embodiments, the button **22** could be located on a side of housing **11**.
(36) Other manual or automated features can include drug injection or needle retraction, or both. Injection is the process by which a bung or piston **23** is moved from a proximal location within a

syringe (not shown in FIGS. **1**A and **1**B) to a more distal location within the syringe in order to force a medicament **15** from the syringe through needle **17**. In some embodiments, a biasing member such as a drive spring (not shown in FIGS. **1**A and **1**B) is under compression before device **10** is activated. A proximal end of the drive spring can be fixed within proximal region **21** of housing **11**, and a distal end of the drive spring can be configured to apply a compressive force to a proximal surface of piston **23**. Following activation, at least part of the energy stored in the drive spring can be applied to the proximal surface of piston **23**. This compressive force can act on piston **23** to move it in a distal direction. Such distal movement acts to compress the liquid medicament **15** within the syringe, forcing it out of needle **17**.
(37) Following injection, needle **17** can be retracted within the needle sleeve **13** or housing **11**. Retraction can occur when the needle sleeve **13** moves distally as a user removes device **10** from a subject's body. This can occur as needle **17** remains fixedly located relative to housing **11**. Once a distal end of sleeve **13** has moved past a distal end of needle **17**, and needle **17** is covered, sleeve **13** can be locked. Such locking can include locking any proximal movement of sleeve **13** relative to housing **11**.
(38) Another form of needle retraction can occur if needle **17** is moved relative to housing **11**. Such movement can occur if the syringe within housing **11** is moved in a proximal direction relative to housing **11**. This proximal movement can be achieved by using a retraction spring (not shown), located in distal region **20**. A compressed retraction spring, when activated, can supply sufficient force to the syringe to move it in a proximal direction. Following sufficient retraction, any relative movement between needle **17** and housing **11** can be locked with a locking mechanism. In addition, button **22** or other components of device **10** can be locked as required.
(39) FIG. **2** shows a simplified view of a medicament delivery device **100** that extends along an axis **144**. The medicament delivery device **100** may share one or more features with the drug delivery device **10** discussed in relation to FIGS. **1**A and **1**B.
(40) The medicament delivery device **100** may be configured to inject greater than 2 ml of medicament and/or the medicament delivery device **100** may be configured to inject medicament having a viscosity of greater than 25 cP. Nevertheless, in other examples the medicament delivery device **100** may be configured to inject 2 ml or less of medicament and/or the medicament delivery device **100** may be configured to inject medicament having a viscosity of 25 cP or less.
(41) The medicament delivery device **100** has a body **111** having a proximal end **130** and a distal end **131** arranged along the axis **144**, a hollow needle **117** for injecting medicament **115**, and a needle cover **113**. The body **111** is shown to be substantially cylindrical, however it should be understood that the body **111** may have a different shape in other examples.
(42) The body **111** houses a pre-filled syringe **150**, which comprises a container **114** containing the medicament **115**. The needle **117** is coupled to a distal end of the container **114** and is in fluid communication with an interior of the container **114** such that the medicament **115** may be dispensed from the container **114**, via the needle **117**. FIG. **2** shows the needle **117** permanently coupled to the container **114**, however it should be understood that this is not meant to be limiting. For example, in other instances, the needle **117** may be removably couplable to the container **114** (e.g., via a Luer lock interface between a connector on the container **114** and a connector coupled to the needle **117**) such that the needle **117** may be coupled to the container **114** prior to an injection and uncoupled from the container **114** after an injection (e.g., to replace the needle **117** with a new needle **117**).
(43) The syringe **150** further comprises a bung or piston **123** arranged within the container **114**, proximal to the medicament **115** and the needle **117**. The piston **123** is arranged within the container **114** to be moved distally to force the medicament **115** out of the container **114**, via the needle **117**.
(44) The needle **117** has a distal end **140**. The needle cover **113** is proximally movable relative to the body **111** between an extended position, in which the needle cover **113** covers the distal end **140**

of the needle **117**, and a retracted position, in which the distal end **140** of the needle **117** protrudes from the needle cover **113** for penetration into an injection site. FIG. **2** shows the device **100** with the needle cover **113** in the extended position. The medicament delivery device **100** further comprises a needle cover biasing member **118**, such as a spring, configured to bias the needle cover **113** axially in the distal direction. The distal direction is indicated by the direction of the arrow **142** in FIG. **2**.

(45) The medicament delivery device **100** comprises a medicament delivery mechanism **180** for dispensing the medicament **115** from the syringe **150** held within the body **111**. The medicament delivery mechanism **180** comprises a plunger **121**, a collar **119** and a drive member **124**.

(46) The plunger **121**, which may be coaxial with the axis **144**, is axially movable within the syringe **150** of the device **100** in a distal direction to dispense the medicament **115** from the syringe **150** via the needle **117**. The plunger **121** is arranged to engage the piston **123** of the syringe **150** such that distal axial movement of the plunger **121** moves the piston **123** distally to dispense the medicament **115** via the needle **117**.

(47) The collar **119**, which may also be coaxial with the axis **144**, is axially fixed relative to the body **111** but is rotatable within the body **111** (e.g., about the axis **144**). The drive member **124** is configured to rotate the collar **119** when the drive member **124** is actuated/released. For example, the drive member **124** may be a spring (e.g., a torsion spring), wherein the spring is configured to rotate the collar **119** when the spring is released. However, it should be understood that one or more other types of drive member **124** may be used instead, such as a pneumatic drive member. The drive member **124** is released when the needle cover **113** reaches a predetermined axial displacement with respect to the body **111**.

(48) The collar **119** and the plunger **121** are arranged such that the rotation of the collar **119** by the drive member **124** causes the plunger **121** to move distally within the syringe **150**, to thereby dispense the medicament **115** from the syringe **150** via the needle **117**. For example, the plunger **121** may comprise an external screw thread **122** that is configured to interface with an internal screw thread of the collar **119** such that rotation of the collar **119** causes distal translation of the plunger **121**. However, it should be understood that other forms of interface between the collar **119** and the plunger **121** for converting rotation of the collar **119** into translation of the plunger **121** may be used instead. For example, in other instances, only one of the collar **119** and the plunger **121** may have a screw thread, and the other of the collar **119** and the plunger **121** may have one or more engagement features, such as one or more projections, that are arranged to engage with the screw thread such that rotation of the collar **119** causes translation of the plunger **121**.

(49) To initiate delivery of the medicament **115** into a subject (who may be the user of the medicament delivery device **100**, a different person to the user of the device, or an non-human animal), a distal end **120** of the needle cover **113** is to be pressed against the injection site on the subject and the body **111** is moved towards the injection site, thereby moving the needle cover **113** axially into the body **111** and uncovering the needle **117** from within the needle cover **113** such that it penetrates the injection site. The proximal axial displacement of the needle cover **113** causes the release of the drive member **124**, which rotates the collar **119**. The rotation of the collar **119** moves the plunger **121** axially within the syringe **150** to dispense the medicament **115** into the injection site via the needle **117**. The device **100** is pressed against the injection site to hold the needle cover **113** in its retracted position whilst the medicament **115** is dispensed from the device **100**.

(50) After the medicament **115** has been dispensed, the device **100** is removed from the injection site by moving the body **111** away from the injection site. In doing so, the needle cover **113** moves distally under the force of the biasing member **118** towards the extended position to cover the distal end **140** of the needle **117** and therefore protect the user and/or subject from an accidental needle-stick event. In some instances, subsequent proximal movement of the needle cover **113** relative to the body **111** may be inhibited by a locking mechanism.

(51) FIGS. **3**A to **3**F show a simplified cross-section view of a medicament delivery device **200** in

various stages of operation, in accordance with one or more aspects of the present disclosure.

(52) The features described and/or contemplated in relation to the medicament delivery device **200** may be incorporated in the medicament delivery device **100** described and/or contemplated above in relation to FIG. **2**. Alternatively or additionally, the features described and/or contemplated in relation to the medicament delivery device **200** may be incorporated in another medicament delivery device, for example a medicament delivery device having a different mechanism for dispensing a medicament to that described in relation to the medicament delivery device **100**. Like references refer to like features.

(53) With reference to FIG. **3**A, the medicament delivery device **200** includes a body **111** extending along an axis **144** of the medicament delivery device **200** and having a proximal end **130** and a distal end **131**. A syringe **150** containing a medicament **115** is held within the body **111**.

(54) The medicament delivery device **200** further includes a needle **117** for injecting the medicament **115** and a needle cover **113** axially movable relative to the body **111** between an extended position and a retracted position. When in the extended position, the needle cover **113** extends from the distal end **131** of the body **111** such that a distal end **140** of the needle **117** is surrounded, to protect a user from an accidental needle-stick injury prior to an actual injection. When the needle cover **113** is in the retracted position, the distal end **140** of the needle **117** protrudes distally from the distal end **120** of the needle cover **113** such that the distal end **140** of the needle **117** can penetrate an injection site. A needle cover biasing member **118**, which in this example takes the form of a spring, is configured to bias the needle cover **113** distally, from the retracted position to the extended position. It should be understood that, in other examples, the needle cover biasing member **118** may take a different form to a spring, for example a pneumatic mechanism or an elastic polymer.

(55) In FIG. **3**A, the medicament delivery device **200** is shown in an initial state prior to an injection, in which the medicament delivery device **200** is not being held against an injection site. The needle cover **113** is in the extended position with respect to the body **111**, biased into the extended position by the needle cover biasing member **118**. The needle cover **113** is axially movable relative to the body **111** between the extended position shown in FIG. **3**A, in which a distal end **120** of the needle cover **113** is distal to a distal end **140** of the needle **117**, and the retracted position shown in FIG. **3**B, in which the distal end **140** of the needle **117** is distal to the distal end **120** of the needle cover **113**.

(56) The medicament delivery device **200** comprises a medicament delivery mechanism **180** for dispensing the medicament **115** from the syringe **150** held within the body **111**. The medicament delivery mechanism **180** in this example comprises a plunger **121**, a collar **119** and a drive member such as a spring (e.g., torsion spring), one or more of which may be as described above in relation to the device **100**. However, it should be understood that in other examples, the medicament delivery mechanism **180** may comprise one or more different components than a plunger **121**, a collar **119** and/or drive member **124** described above in relation to the device **100**.

(57) The plunger **121**, which may be coaxial with the axis **144**, is axially movable within the syringe **150** of the medicament delivery device to dispense medicament **115** from the syringe **150** via the needle **117**. The plunger **121** is located proximal to the piston **123** of the syringe **150**. FIG. **3**A shows that the distal end of the plunger **121** is initially axially separated from the piston **123** when the medicament delivery device **200** is in its initial state. However, it should be understood that in other examples the distal end of the plunger **121** may be engaged with the piston **123** when the medicament delivery device **200** is in its initial state.

(58) The collar **119**, which may be coaxial with the axis **144**, is axially fixed relative to the body **111** but is able to rotate with respect to the body **111** (e.g., about the axis **144**). The drive member (which is not shown in FIGS. **3**A-**3**F to improve the clarity of the drawings, but which may be similar or identical in location and/or type to the drive member **124** previously described in relation to FIG. **2**) is configured to rotate the collar **119** when released. The drive member may be a spring

such as a torsion spring, however other forms of spring/drive member may be used instead.

(59) The drive member (e.g., spring) is released when the needle cover **113** reaches a predetermined axial displacement relative to the body **111**. The collar **119** interfaces with the plunger **121** (e.g. via any interface(s) previously described in relation to FIG. **2**) such that the rotation of the collar **119** by the drive member causes the plunger **121** to move distally within the syringe **150** to thereby dispense medicament **115** from the syringe **150** via the needle **117** (e.g., in the same or similar manner as previously described in relation to FIG. **2**).

(60) The medicament delivery device **200** has a latch **160** that is releasably engageable with the medicament delivery mechanism **180** for initiating, pausing and resuming dispensing of the medicament **115** by the medicament delivery mechanism **180**. The latch **160** is movable by the needle cover **113** between an engaged configuration, in which the latch **160** is engaged with a component of the medicament delivery mechanism **180** to prevent delivery of the medicament **115**, and a disengaged configuration, in which the latch **160** is disengaged from the component of the medicament delivery mechanism **180** to allow (e.g. initiate and/or resume) delivery of the medicament **115**. The latch **160** comprises an engaging element **161** configured to engage the component of the medicament delivery mechanism **180** when in the engaged configuration and to disengage the component of the medicament delivery mechanism **180** when in the disengaged configuration.

(61) FIG. **3**A shows the latch **160** in the engaged configuration. In this example, the component of the medicament delivery mechanism **180** that is engaged and disengaged by the latch **160** is the collar **119**, as later described in more detail in relation to FIGS. **4**A-**4**C. However, it should be understood that, in other examples, the latch **160** may be configured to engage a different component than the collar **119** to initiate, pause and/or resume medicament delivery. For example, the latch **160** may additionally/alternatively be configured to engage and disengage the plunger **121** and/or drive member to initiate, pause and/or resume medicament delivery.

(62) FIGS. **4**A-**4**C each show a schematic cross-section of a portion of the medicament delivery device **200** to demonstrate operation of the latch **160**, wherein the cross-section is in a plane that is normal to the axis **144**.

(63) FIG. **4**A shows a cross-section through portions of the needle cover **113**, latch **160**, collar **119** and plunger **121** of the medicament delivery device **200** of FIG. **3**A when the medicament delivery device **200** is in its initial state, with the latch **160** in its engaged configuration. The view in FIG. **4**A is along the axis **144** of the medicament delivery device **200** (i.e., the axis **144** is normal to the page). The plunger **121** is arranged along the axis **144**. The collar **119** is arranged radial to the plunger **121**, such that it surrounds at least a portion of the plunger **121**. The needle cover **113** is arranged radial to the collar **119** and plunger **121**, such that it surrounds at least a portion of the collar **119** and plunger **121**. The plunger **121**, collar **119** and needle cover **113** may be concentric about the axis **144**.

(64) The collar **119** is biased to rotate about the axis **144** by the drive member for causing distal translation of the plunger **121**, however rotation of the collar **119** is limited (e.g., prevented) while the medicament delivery device **200** is in the initial state by engagement of the latch **160** with the collar **119**. The latch **160** comprises a flexible arm **162** that extends at least partially around the collar **119**, between an outer surface of the collar **119** and an inner surface the needle cover **113**. An engaging element **161**, for example in the form of a protrusion, extends radially inwards from a free end of the flexible arm **162** to releasably engage with a corresponding engaging element **191** of the collar **119**, in this example the engaging element **191** of the collar **119** being a recess on the outer surface of the collar **119**. FIG. **4**A shows the engaging element **161** of the latch **160** engaged with the engaging element **191** of the collar **119** such that rotation of the collar **119** relative to the latch **160** is limited. The latch **160** may be coupled to the body **111** such that engagement between the engaging element **161** of the latch **160** and the engaging element **191** of the collar **119** limits (e.g., prevents) rotation of the collar **119** relative to the body **111**. As such, the medicament delivery

mechanism **180** is inhibited from dispensing medicament **115** when engaged by the latch **160**.

(65) The latch **160** may be biased from the engaged configuration to the disengaged configuration. In FIG. **4**A, the latch **160** is biased from the engaged configuration towards the disengaged configuration at least in part by the medicament delivery mechanism **180**. More specifically, the collar **119** is biased to rotate by the drive member of the medicament delivery mechanism **180**, wherein this biased rotation biases the engaging element **161** of the latch **160** out of engagement with the engaging element **191** of the collar **190**, for example due to engagement between a side wall **166** of the engaging element **161** of the latch **160** and a side wall **196** of the engaging element **191** of the collar **119**. In some examples, one or both of the side walls **166**, **196** may be ramped (e.g., bevelled) to assist with biasing the engaging element **161** of the latch **160** to disengage the engaging element **191** of the collar **119**. Additionally or alternatively, in some examples the latch **160** may be biased from the engaged configuration to the disengaged configuration at least in part by a resilient portion of the latch **160**. For example, at least a portion of the flexible arm **162** may be formed of a resilient material that causes the free end of the flexible arm **162** to be biased radially outwards, to disengage the engaging element **161** of the latch **160** from the engaging element **191** of the collar **119**. Other suitable mechanisms for biasing the latch **160** towards its disengaged configuration may be envisaged.

(66) As shown in FIG. **4**A, the latch **160** is held in the engaged configuration by an inner surface of the needle cover **113** when the needle cover **113** is in the extended position. As such, the free end of the flexible arm **162** is inhibited from moving radially away from the collar **119** and so the engaging element **161** (e.g. protrusion) of the latch **160** is held in engagement with the engaging element **191** of the collar **119** to limit rotation of the collar **119** while the needle cover **113** is in the extended position.

(67) The needle cover **113** is configured such that proximal movement of the needle cover **113** from the extended position towards the retracted position moves the latch **160** from the engaged configuration to the disengaged configuration, as later described in relation to FIGS. **3**B and **4**B, while distal movement of the needle cover **113** from the retracted position towards the extended position moves the latch **160** from the disengaged configuration to the engaged configuration, as later described in relation to FIGS. **3**C and **4**C.

(68) FIG. **4**A shows the collar **119** may have a plurality of engaging elements **191** (e.g., recesses) arranged circumferentially around an outer surface of the collar **119** such that the engaging element **161** of the latch **160** remains aligned with one of those plurality of engaging elements **191** for engagement, regardless of the rotational position of the collar **119**.

(69) As shown in FIG. **3**A, the medicament delivery device **200** further comprises a needle cover lock **600** for limiting proximal movement of the needle cover **113** after the completion of medicament delivery. FIG. **3**A shows the needle cover lock **600** in an initial configuration, in which movement of the needle cover **113** from the extended position to the retracted position is not limited by the needle cover lock **600**, therefore the needle cover **113** is free to move proximally. The needle cover lock **600** may be movable by the medicament delivery mechanism **180** between the initial configuration and a locking configuration, wherein movement of the needle cover **113** from the extended position to the retracted position is limited by the needle cover lock **600** when in the locking configuration. In some examples, the needle cover lock **600** may be moved from the disengaged position to the engaged position by distal movement of the plunger **121** of the medicament delivery mechanism **180**. Operation of the needle cover lock **600** is described in more detail later, in relation to FIGS. **6**A-**6**D.

(70) As shown in FIG. **3**A, the needle cover **113** of the medicament delivery device **200** comprises a pair of arms **126**. The arms **126** extend in a proximal direction from the remainder of the needle cover **113**. It should be understood that in other examples the needle cover **113** may comprise a different number of arms **126** to two, for example the needle cover **113** may comprise a single arm **126** or more than two arms **126** (e.g., three arms **126** or four arms **126** etc.).

(71) The or each arm **126** has a protrusion **127**. The protrusion **127** is at the free (e.g., proximal) end of the or each arm **126**, although in other embodiments the protrusion **127** may be located distally from the free end of the arm(s) **126**. The arm(s) **126** and/or protrusion(s) **127** may be flexible, as discussed later.
(72) FIG. **3**A shows the pair of arms **126** and respective protrusions **127** arranged symmetrically about the axis **144** (e.g., with a rotational symmetry of 180 degrees about the axis **144**). However, it should be understood that this is not meant to be limiting and that other arrangements are possible. For example, the pair of arms **126** may be separated by an angle of 90 degrees about the axis **144**.
(73) The medicament delivery device **200** has a needle cover guide **143**, which may be arranged at least partially within the body **111**. The needle cover guide **143** comprises a track **128** arranged on its outer circumferential surface. In some examples a separate track **128** is provided for each arm **126** (if more than one arm **126** is present), or a single track **128** may be used for each arm **126**.
(74) The medicament delivery device **200** further comprises an actuation member **170** configured to engage the needle cover guide **143**, as described later in relation to FIGS. **5**A-**5**E. FIG. **3**A shows the actuation member **170** taking the form of a button arranged at a proximal end of the medicament delivery device **200**. However, it should be understood that the actuation member **170** may, in other examples, take a different form to a button, and/or be located at a different location of the medicament delivery device **200** and/or body **111**.
(75) The actuation member **170** is actuatable by a user between a first position and a second position relative to the body **111**. FIG. **3**A shows the actuation member **170** in the first position relative to the body **111**, while FIG. **3**D shows the actuation member **170** in the second position relative to the body **111**, the actuation member **170** having been moved distally relative to the body **111** from the first position to the second position by actuation (e.g., movement in the distal direction) of the actuation member **170** by a user. A user may directly press the actuation member **170** to move it distally between the first position and the second position, for example by pushing a proximally-facing surface of the actuation member **170** distally with a finger or thumb. The actuation member **170** may be biased from its second position to its first position by an actuation member biasing member such as a spring **171**, however other forms of actuation member biasing member to a spring **171** may be envisaged.
(76) The or each arm **126** engages the track **128** of the needle cover guide **143** via the protrusion(s) **127**. The track **128** is configured to limit a proximal movement of the needle cover **113** relative to the body **111**, after the needle cover **113** has moved from the retracted position back to the extended position. That is, the or each arm **126** is configured to engage the track **128** such that a proximal movement of the needle cover **113** is limited after the needle cover **113** has moved from the retracted position to the extended position.
(77) The needle cover guide **143** is substantially cylindrical and is rotatable relative to the body **111**, for example about the axis **144**, but is axially fixed relative to the body **111**. FIG. **3**A shows the needle cover guide **143** arranged at a proximal end of the medicament delivery device **200**, however it should be understood that in other examples the needle cover guide **143** may be located at a different location of the medicament delivery device **200**, for example distal to the proximal end of the medicament delivery device **200**.
(78) The needle cover guide **143**, track **128** and actuation member **170** are shown in greater detail in FIGS. **5**A-**5**E.
(79) FIGS. **5**A-**5**E show the needle cover guide **143**, track **128** and actuation member **170** of the medicament delivery device **200** of FIGS. **3**A-**3**F in various operational states. In each of FIGS. **5**A-**5**E, only a portion of (e.g., a proximal end of) the arms **126** is shown. Portions of the actuation member **170** are shown as see-through, so that interaction between the actuation member **170** and the needle cover guide **143** is visible. Various other features of the medicament delivery device **200** such as the spring **171** and body **111** are hidden in FIGS. **5**A-**5**E, for clarity.
(80) FIG. **5**A shows the needle cover guide **143**, needle cover **113** and actuation member **170** in a

first configuration, corresponding to their configuration in FIG. **3**A. As shown in FIG. **5**A, the track **128** extends at least partially around a circumference of the needle cover guide **143** for rotating the needle cover guide **143** as the needle cover **113** moves from the extended position to the retracted position.

(81) As shown in FIG. **5**A, the arms **126** are positioned radially outwardly of the needle cover guide **143**, with the protrusion **127** extending radially inwards to engage the track **128** arranged at the outer circumferential surface of the needle cover guide **143**. However, it should be understood that in one or more other embodiments, the arms **126** are instead positioned radially inwardly of the needle cover guide **143**, with the protrusions **127** extending radially outwards to engage the track **128** arranged at an inner circumferential surface of the needle cover guide **143**.

(82) The track **128** comprises a first region **511**, a second region **512**, a third region **513** and a fourth region **514**. When the medicament delivery device **200** is in its initial state shown in FIG. **3**A, with the needle cover **113** in the extended position, the or each arm **126** engages the track **128** via its protrusion **127** at a respective first position **501** within the first region **511** (see FIG. **5**A). The first position **501** may be located near a distal end of the needle cover guide **143**, at a distal end of the first region **511**.

(83) As shown in FIG. **5**A, the first region **511** extends at least partially around the circumference of the needle cover guide **143**. The first region **511** and the second region **512** are arranged such that the protrusion **127** travels from the first position **501** within the first region **511** to a second position **502** within the second region **512** as the needle cover **113** moves (e.g., as indicated by arrow **521**) from the extended position to the retracted position (as later described in relation to FIG. **5**B). The first region **511** has an axially-extending component of direction and is at an acute angle relative to the axis **144** of the medicament delivery device **200**, extending helically about the outer surface of the needle cover guide **143**. The needle cover guide **143** therefore rotates (e.g., as indicated by arrow **520**) when the protrusion **127** travels along the first region **132** of the track **128**, due to engagement between the protrusion **127** and an angled proximal side wall **530** of the first region **501**.

(84) The second region **512** of the track **128** extends axially along the needle cover guide **143** such that the protrusion **127** travels distally within the second region **512** as the needle cover **113** moves distally, from its retracted position to its extended position, without causing the needle cover guide **143** to be rotated.

(85) The third region **513** is located distal to the second region **512** but axially aligned with the second region **512**, such that the protrusion **127** travels distally from the second position **502** in the second region **512** to a third position **503** in the third region **513** as the needle cover **113** moves distally from the retracted position to the extended position, as described later.

(86) The track **128** comprises a locking element **540** arranged at a distal end of the second region **512**, proximal of the third region **513**. The locking element **540** is configured to limit proximal movement of the needle cover **113** after the needle cover **113** has moved from the retracted position to the extended position (and therefore after the protrusion **127** has moved from the second position **502** to the third position **503**). The locking element **540** comprises a proximally-facing ramped surface **541** and a distally-facing locking surface **542** which may be substantially perpendicular to the axis **144**. Operation of the locking element **540** shall be described later.

(87) The third region **513** may have a side wall **543** to limit back-rotation (e.g., rotation in a direction opposite to the rotation indicated by arrow **520**) of the needle cover guide **143** once the protrusion **127** is in the third position **503**.

(88) The third region **513** and the fourth region **514** extend circumferentially around the needle cover guide **143** and are arranged adjacent to each other such that, when the protrusion **127** is in the third position **503** of the third region **513** and the needle cover guide **143** is rotated relative to the needle cover **113** in the direction indicated by the arrow **520**, the protrusion travels from the third position **503** in the third region **513** to a fourth position **504** in the fourth region **514**.

(89) The fourth region **514** may be substantially the same as the first region **511** (e.g., same features etc.), but located at a different circumferential position about the outer circumferential surface of the needle cover guide **143**. That is, the track **128** may comprise a plurality of portions **560** arranged circumferentially around the needle cover guide **143**, each portion **560** comprising a respective first region **511**, second region **512** and third region **513**, wherein each first region **511**, second region **512** and third region **513** may be substantially similar or identical. Each first region **511**, second region **512** and third region **513** of each portion **560** may be spaced about the circumferential outer surface of the needle cover guide **154** such that a first position **501** and first region **511** of one portion **560** acts as the fourth position **504** and fourth region **514** of an adjacent portion **560** of the track **128**. The plurality of portions **560** may be repeated around the entire outer circumferential surface of the needle cover guide **143** such that all of the portions **560** are linked, allowing the protrusion **127** to travel through the first to fourth regions **511**, **512**, **513**, **514** of each portion **560** in order, in a continuous loop around the needle cover guide **143**.
(90) FIGS. **5**A-**5**D show the track **128** comprising four identical portions **560***a*, **560***b*, **560***c*, **560***d* arranged circumferentially around the needle cover guide **143**, each portion **560***a*, **560***b*, **560***c*, **560***d* comprising a respective first region **511**, second region **512** and third region **513**. The four portions **560***a*, **560***b*, **560**, **560***d* are spaced 90 degrees apart around the circumference of the needle cover guide **143**, such that the needle cover guide **143** has a rotational symmetry of four about the axis **144**. While one of the arms **126** engages with a first portion **560***a* of the track, the other of the arms **126** simultaneously engages with a different portion of the track **128** (e.g., the third portion **560***c*) such that movements of a protrusion **127** in the first portion **560***a* of the track **128** mirror movements of the other protrusion **127** in the other portion (e.g., the third portion **560***c*) of the track **128**.
(91) It should be understood that in other examples, there may be a different number of identical portions **560***a-d* of the track **128** to four located around the needle cover guide **143**. For example, there may be two identical portions **560**, three identical portions **560**, or five or more identical portions **560**, each portion **560** having first to fourth regions **511**, **512**, **513**, **514**. In other examples, only a single portion **560** may be present, with the portion **560** extending around the entire circumference of the needle cover guide **143** such that the first position **501** and first region **511** of the portion **560** also act as the fourth position **504** and fourth region **514**.
(92) FIG. **5**A also shows the actuation member **170**, which has four ramped engagement elements **580***a*-**580***d* extending distally from a distal-facing inner surface of the actuation member **170**. Each engagement element **580***a*-**580***d* is configured to engage a respective one of four corresponding ramped engagement elements **581***a*-**581***d* of the needle cover guide **143**, the engagement elements **581***a*-**581***d* extending proximally from a proximal-facing surface of the needle cover guide **143**. The four engagement elements **580***a*-**580***d* of the actuation member are arranged in a circular pattern about the axis **144**, as are the four corresponding engagement elements **581**-**581***d* of the needle cover guide **143**. FIG. **5**A shows the actuation member **170** biased in its first position.
(93) The four engagement elements **580***a*-**580***d* of the actuation member and the four corresponding engagement elements **581**-**581***d* of the needle cover guide **143** are currently arranged such that a distal actuation of the actuation member **170** towards the needle cover guide **143**, from its first position to its second position, does not cause the needle cover guide **143** to substantially rotate. Each of the four engagement elements **580***a*-**580***d* has a corresponding distally-facing ramped surface **582***a*-**582***d*, while each of the four engagement elements **581**-**581***d* has a corresponding proximally-facing ramped surface **583***a*-**583***d*. The distally-facing ramped surfaces **582***a*-**582***d* are configured to engage the proximally-facing ramped surfaces **583***a*-**583***d*, as described later.
(94) While FIG. **5**A shows four engagement elements **580***a*-**580***d* and four corresponding engagement elements **581**-**581***d*, it should be noted that in other cases fewer or more than four engagement elements **580***a*-**580***d* and/or engagement elements **581**-**581***d* may be present.
(95) FIG. **3**B shows the medicament delivery device **200** of FIG. **3**A in a holding state, in which the

needle cover **113** has been moved axially in a proximal direction with respect to the body **111**. The needle cover **113** is shown in its retracted position, in which the distal end **140** of the needle **117** extends distal to the distal end **120** of the needle cover **113**. The user has placed the distal end **120** of the needle cover **113** against an injection site and moved the body **111** towards the injection site, causing the needle cover **113** to move proximally towards its retracted position (shown in FIG. **3**B) and allowing the distal end **140** of the needle **117** to penetrate an injection site of the subject (wherein the subject may be the user, or a different person to the user).

(96) As the needle cover **113** is moved proximally, the latch **160** is moved from its engaged configuration to its disengaged configuration. As such, the latch **160** will be brought out of engagement with the component (e.g., collar **119**) of the medicament delivery mechanism **180**, thereby allowing the medicament **115** to be dispensed.

(97) FIG. **3**B shows that proximal movement of the needle cover **113** has brought a recess **301** in the needle cover **113** into alignment with the latch **160** such that the latch **160** moves from its engaged configuration to its disengaged configuration with respect to the collar **119**.

(98) FIG. **4**B shows the latch **160** in its disengaged configuration, corresponding to FIG. **3**B. The needle cover **113** has been moved proximally (e.g., parallel to the axis **144**, out of the page) from the extended position to the retracted position, bringing the needle cover recess **301** into alignment with the flexible arm **162** of the latch **160**. The flexible arm **162** is no longer held in its engaged position by the needle cover **113**, therefore the flexible arm **162** can now flex radially outwards, into the needle cover recess **301**, due to the biasing of the flexible arm **162** from its engaged configuration to its disengaged configuration. Radial movement of the flexible arm **162** outwards brings the engaging element **161** (e.g., projection) of the latch **160** out of engagement with the engaging element **191** (e.g., recess) of the collar **119** such that the latch **160** is now in its disengaged configuration and rotation of the collar **119** by the drive member is no longer limited by the latch **160**.

(99) Returning to FIG. **3**B, it can be seen that disengagement of the latch **160** from the collar **119** has initiated medicament delivery. Disengagement of the latch **160** from the collar **119** has allowed the collar **119** to be rotated by the drive member and, through interaction between the collar **119** and the plunger **121**, cause the plunger **121** to move axially in a distal direction to engage the piston **123** (if not already engaged) and move the piston **123** distally to dispense at least a portion of the medicament **115** into the injection site. The delivery of medicament **115** is not yet complete, therefore the needle cover lock **600** remains in its in its initial configuration in which proximal movement of the needle cover **113** is not limited by the needle cover lock **600**.

(100) It is also shown in FIG. **3**B that proximal movement of the needle cover **113** from the extended position to the retracted position has rotated the needle cover guide **143**, due to the engagement between the arms(s) **126** and the track **128**. Proximal movement of the needle cover **113** and rotation of the needle cover guide **143** has moved the locking element(s) **540** of the needle cover guide **143** relative to the projection(s) **127** such that it is (they are) now distal to the projection(s) **127** and axially aligned relative to the projection(s) **127**.

(101) FIG. **5**B shows the needle cover guide **143**, needle cover **113** and actuation member **170** in a second configuration, corresponding to their configuration in FIG. **3**B. As the needle cover **113** has moved axially from the extended position to the retracted position (i.e., ‘upwards’ between FIG. **5**A and FIG. **5**B, as indicated by arrow **521**), the projection(s) **127** of the flexible arm(s) **126** has travelled from the first position **501**, in the first region **511** of the track **128**, to the second position **502**, in the second region **512** of the track **128**, causing the needle cover guide **143** to rotate (e.g., as indicated by the arrow **520**) relative to the arm(s) **126**. The second position **502** is at a proximal end of the second region **512**, with the second position **502** proximal to the first position **501** and spaced from the first position **501** circumferentially around the circumferential outer surface of the needle cover guide **143**.

(102) FIG. **5**B shows the actuation member **170** remains in its first position and has not rotated

relative to the needle cover **113**. However, the needle cover guide **143** has rotated relative to the actuation member **170** such that the engagement elements **581***a*-**581***d* of the needle cover guide **143** have been rotated relative to the engagement elements **580***a*-**580***d* of the actuation member **170**, bringing them into a new alignment with the engagement elements **580***a*-**580***d*. The engagement elements **581***a*-**581***d* of the needle cover guide **143** and the engagement elements **580***a*-**580***d* of the needle cover guide **143** are now in a relative configuration such that a distal movement of the actuation member **170** from its first position to its second position would cause the needle cover guide **143** to be rotated relative to the actuation member **170** and needle cover **113** in the direction indicated by arrow **520**, as described later.
(103) Once the medicament delivery process has been initiated, the drive member of the medicament delivery mechanism **180** will continue to automatically dispense the medicament **115**. However, the user of the medicament delivery device **200** may wish to pause the medicament delivery process after it has been initiated, before the entire medicament **115** has been dispensed.
(104) A user may wish to pause a medicament delivery process after only a portion of the medicament **115** has been delivered to a subject such that one or more remaining portions of the medicament **115** may be delivered to the same subject (or another subject) at a later time. The user may wish to deliver the medicament **115** in a plurality of doses because the subject is experiencing discomfort during the medicament delivery process, for example due to the volume of the medicament **115** being delivered and/or the rate at which the medicament **115** is being delivered. Additionally or alternatively, the user may wish to deliver the medicament **115** in a plurality of discrete doses spaced apart in time because the dosage regimen of the medicament **115** requires such a method of administration (e.g., the dosage regimen requires three doses to be delivered, each separated by one hour). Additionally or alternatively, the user may have inserted the needle **117** of the medicament delivery device **200** at an incorrect injection site and would like to pause the medicament delivery process so that the needle **117** may be removed and repositioned at a correct injection site. It should be understood that this list of reasons for pausing the medicament delivery process is not exhaustive and that there may be other reasons for a user desiring to pause the medicament delivery process after it has been initiated.
(105) One or more aspects of the present disclosure allow the medicament delivery process to be paused and resumed by a user.
(106) To pause the medicament delivery process after it has been initiated, the user may remove the medicament delivery device **200** from the injection site. FIG. **3**C shows the medicament delivery device **200** of FIG. **3**B in a paused state, after it has been removed from the injection site. The user has moved the body **111** away from the injection site, causing the needle cover **113** to translate axially with respect to the body **111** in a distal direction due to the biasing force applied by the needle cover biasing member **118**.
(107) Distal movement of the needle cover **113** relative to the body **111** from the retracted position shown in FIG. **3**B towards the extended position shown in FIG. **3**C has paused the medicament delivery process. That is, distal movement of the needle cover **113** from the retracted position towards the extended position has moved the latch **160** from the disengaged configuration shown in FIG. **3**B, in which delivery of the medicament **115** by the medicament delivery mechanism **180** is allowed, to the engaged configuration shown in FIG. **3**C, in which delivery of the medicament **115** by the medicament delivery mechanism **180** is inhibited (e.g., prevented).
(108) FIG. **4**C shows a cross-section through portions of the needle cover **113**, latch **160**, collar **119** and plunger **121** of the medicament delivery device **200** when the medicament delivery device **200** is in its paused state shown in FIG. **3**C. Distal movement of the needle cover **113** (i.e., along the axis **144**, into the page) has brought the needle cover recess **301** out of alignment with the latch **160** to cause the latch **160** to move from its disengaged configuration shown in FIG. **4**B to its engaged configuration shown in FIG. **4**C. As the needle cover **113** has moved distally, the inner surface of the needle cover **113** has been bought back into contact with the flexible arm **162**, urging the

flexible arm **162** to move radially inwards such that the engaging element **161** (e.g., protrusion) engages the collar **119**.
(109) The needle cover recess **301** may have a ramped side wall **302** (shown in FIGS. **3**B and **3**C) located at a proximal end of the recess **301** that is configured to engage the flexible arm **162** as the needle cover **113** moves distally, to assist with moving the flexible arm **162** radially inwards.
(110) Engagement of the latch **160** with the collar **119** prevents further rotation of the collar **119** under the bias of the drive member, thereby pausing delivery of the medicament **115**. As shown in FIG. **4**C, radial movement of the flexible arm **162** has caused the engaging element **161** (e.g., protrusion) of the latch **160** to engage the engaging element **192** of the collar **119** to prevent rotation of the collar **119**. The engaging element **192** of the collar **119** may be the same engaging element **192** previously discussed in relation to FIG. **4**A with which the engaging element **161** of the latch **160** was previously engaged, or it may be a different engaging element **192** of the collar **119** (e.g., where the collar **119** comprises a plurality of engaging elements **192** arranged around its circumferential outer surface, and the collar **119** is in a different rotational alignment with the latch **160** due to the medicament delivery process).
(111) As the needle cover **113** has moved distally from the retracted position (shown in FIG. **3**B) to the extended position (shown in FIG. **3**C), the projection **127** has travelled distally from the second position **502** in the second region **512** of the track **128** to the third position **503** in the third region of the track **503**. During this movement, the projection **127** has moved over the locking element **540**, with the projection **127** traversing in a distal and radial direction up the ramped surface **541** of the locking element **540** until it passes the locking surface **542**.
(112) FIG. **5**C shows the needle cover guide **143**, needle cover **113** and actuation member **170** in a third configuration, corresponding to their configuration in FIG. **3**C. As the projection **127** has traversed the ramped surface **541** of the locking element **540**, it has been moved radially outwards due to flexibility of the arm **126** and/or projection **127**. Once the projection **127** has moved distally to the locking surface **541**, the projection **127** is no longer engaged by the ramped surface **541** and so now moves radially inwards, due to resiliency of the arm **126** and/or projection **127**. Proximal movement of the needle cover **113** is now limited (i.e., prevented) by engagement between the projection **127** (or a different part of the arm **126**) and the locking surface **542**, as shown in FIG. **5**C. This temporary ‘locking’ of the needle cover **113** to prevent further proximal movement may be beneficial in that it protects the distal end **140** of the needle **117** from being exposed to the user and/or subject after the needle cover **113** has moved back to its extended position, lessening the possibility of an accidental needle-stick event.
(113) Rotation of the needle cover guide **143** relative to the needle cover **113** allows the needle cover **113** to be ‘unlocked’ such that it can be moved proximally again, from its extended position to its retracted position. Such rotation of the needle cover guide **143** may be performed using the actuation member **170**.
(114) FIG. **3**D shows the medicament delivery device **200** of FIG. **3**C in a resetting state, wherein the actuation member **170** has been actuated by a user to rotate the needle cover guide **143** relative to the needle cover **113**. To actuate the actuation member **170**, the user has applied a distal force to a proximally-facing surface of the actuation member **170** (e.g., by pressing the proximally-facing surface with a thumb or finger) to move the actuation member **170** distally, against the biasing force of the spring **171**.
(115) FIG. **5**D shows the needle cover guide **143**, needle cover **113** and actuation member **170** in a third configuration, corresponding to their configuration in FIG. **3**D. Distal axial movement of the actuation member **170** by the user from its first position (shown in FIG. **5**C) to its second position (shown in FIG. **5**D) has caused the needle cover guide **143** to be rotated as indicated by the arrow **520**, due to engagement between the engagement elements **580***a*-**580***d* of the actuation member and the respective engagement elements **581***a*-**581***d* of the needle cover guide **143**. More specifically, as the actuation member **170** has been moved from its first axial position towards its second axial

position, engagement between each of the ramped surfaces **582***a*-**582***d* of the actuation member **170** and corresponding ramped surfaces **583***a*-**583***d* of the needle cover guide **143** has converted the axial movement of the actuation member **170** into rotation of the needle cover guide **143**.
(116) As shown in FIG. **5**D, rotation of the needle cover guide **143** by the actuation member **170** has caused the protrusion **127** to travel from the third position **503** in the third region **513** to the fourth position **504** in the fourth region **514**, such that proximal movement of the needle cover **113** from the extended position to the retracted position is no longer limited by the locking element **540**.
(117) FIG. **5**E shows the needle cover guide **143**, needle cover **113** and actuation member **170** in a fourth configuration, after the user has the user has released the actuation member **170** such that it has moved proximally relative to the needle cover guide **143** from its second position (shown in FIG. **5**D) to its first position (shown in FIG. **5**E), due to the biasing force of the spring **171**. The engagement elements **580***a*-**580***d* of the actuation member **170** have been brought out of engagement with the engagement elements **581***a*-**581***d* of the needle cover guide **143**. Distal axial movement of the actuation member **170** by the user from its first position back to its second position has not caused the needle cover guide **143** to be rotated any further.
(118) As shown in FIG. **3**D, the needle cover lock **600** remains in its initial configuration, in which movement of the needle cover **113** from the extended position to the retracted position is not limited by the needle cover lock **600**. Furthermore, proximal movement of the needle cover **113** from the locked position to the retracted position is no longer limited by the needle cover guide **143** after rotation of the needle cover guide **143** using the actuation member **170**. Therefore, the needle cover **113** is now free to move proximally to resume medicament delivery.
(119) FIG. **3**E shows the medicament delivery device **200** of FIG. **3**E after the medicament delivery process has been resumed.
(120) The user may wish to resume the medicament delivery process to deliver one or more remaining portions of the medicament **115** to the same subject, for example because the subject is no longer experiencing discomfort, because the dosage regimen requires delivery of the medicament in a plurality of discrete doses, and/or because the medicament delivery device **200** has been moved from an incorrect injection site to a correct injection site. Alternatively, the user may wish to resume the medicament delivery process to deliver one or more remaining portions of the medicament **115** to a different subject.
(121) In some examples, if the needle **117** can be uncoupled from the syringe **150**, the user may replace the needle **117** of the medicament delivery device **200** with a new needle **117** prior to resuming the medicament delivery process. This may be performed for reasons of sterility, to prevent cross-contamination, and/or because the previous injection may have blunted the original needle **117**. However, it should be understood that in other examples the needle **117** is not removed or replaced between injections.
(122) FIG. **3**E shows the medicament delivery device **200** of FIG. **3**D now in a resumed state, which may be similar to the holding configuration discussed in relation to FIG. **3**B, however the needle cover guide **143** has been rotated compared to the holding configuration.
(123) As shown in FIG. **3**E, the needle cover **113** has again been moved axially in a proximal direction with respect to the body **111**. The needle cover **113** is shown in its retracted position, in which the distal end **140** of the needle **117** extends distal to the distal end **120** of the needle cover **113**. The user has placed the distal end **120** of the needle cover **113** against an injection site (which may be the same injection site as before, or a different injection site) and moved the body **111** towards the injection site, causing the needle cover **113** to move proximally towards its retracted position and allowing the distal end **140** of the needle **117** to penetrate the injection site.
(124) As the needle cover **113** is moved proximally, the latch **160** has again moved from its engaged configuration to its disengaged configuration, bringing the latch **160** out of engagement with the collar **119** of the medicament delivery mechanism **180** and thereby allowing the dispensing of the remaining medicament **115** to be resumed.

(125) FIG. **3**E shows that proximal movement of the needle cover **113** has again brought the recess **301** in the needle cover **113** into alignment with the latch **160** such that the latch **160** moves from its engaged configuration to its disengaged configuration with respect to the collar **119**. The latch **160** will be in a similar disengaged configuration as previously described in relation to FIG. **4**B. Disengagement of the latch **160** from the collar **119** has allowed the collar **119** to be rotated by the drive member and, through interaction between the collar **119** and the plunger **121**, caused the plunger **121** to again move axially in a distal direction to move the piston **123** distally to dispense at least a portion of the remaining medicament **115** (e.g., in a similar manner as previously described in relation to FIG. **4**B).
(126) It can be seen in FIG. **3**E that movement of the needle cover **113** from the extended position to the retracted position has again caused the needle cover guide **143** to be rotated, in a similar manner as described in relation to FIGS. **5**A and **5**B. Proximal movement of the needle cover **113** and rotation of the needle cover guide **143** has moved the locking element **540** relative to the projection **127** such that it is now distal to the projection **127** and axially aligned relative to the projection **127**. The locking element **540** which may be the same locking element **540** as FIG. **3**C if the track **128** comprises a single portion **560**, or else a different locking element **540** from an adjacent/subsequent portion **560** of the track **128** if the track **128** comprises more than one portion **560**.
(127) The needle cover lock **600** remains in its initial configuration until the medicament delivery process is complete. While the needle cover lock **600** remains in its initial configuration, the user may once again pause the medicament delivery process by removing the medicament delivery device **200** from the injection site and causing the latch **160** to move to its engaged configuration, as described previously. To resume medicament delivery, the user rotates the needle cover guide **143** by actuating the actuation member **170** as described previously, before pressing the medicament delivery device **200** against the injection site to move the latch **160** to its disengaged configuration. The process of pausing and resuming medicament delivery may be repeated multiple times until the medicament delivery process is complete (e.g., after the medicament **115** has been fully dispensed), with the user actuating the actuation member **170** after each pause in the medicament delivery process so that the medicament delivery process can be resumed.
(128) Once the delivery of medicament is complete, the needle cover lock **600** is moved to its locking configuration as shown in FIG. **3**E, wherein movement of the needle cover **113** from the extended position to the retracted position is limited by the needle cover lock **600** when in the locking configuration. When in the locking configuration, a projection **601** of the needle cover lock **600** is able to engage an engaging element **602** of the needle cover **113** to limit proximal movement of the needle cover **113**. FIG. **3**E shows the projection **601** of the needle cover lock **600** has been moved into position by the plunger **121** but is not yet engaged with the engaging element **602** of the needle cover **113** to inhibit proximal movement of the needle cover **113**.
(129) FIG. **3**F shows the medicament delivery device **200** of FIG. **3**E in a locked state after the medicament delivery process is complete.
(130) FIG. **3**F shows the medicament delivery device **200** of FIG. **3**E after the needle cover lock **600** has been moved into its locking configuration by the plunger **121** and the medicament delivery device **200** has been removed from the injection site. The user has moved the body **111** away from the injection site, causing the needle cover **113** to translate axially with respect to the body **111** in a distal direction due to the biasing force applied by the needle cover biasing member **118**. As the needle cover **113** has moved to the extended state, the projection **601** of the needle cover lock **600** has engaged the engaging element **602** of the needle cover **113** to lock the needle cover **113** relative to the body **111** and limit subsequent proximal movement of the needle cover **113**. As such, the medicament delivery device **200** can no longer be used for medicament delivery and can now be disposed of safely.
(131) FIGS. **6**A-**6**D are schematic partial side-views of the medicament delivery device **200** of

FIGS. **3**A-**3**F, showing various operations of the needle cover lock **600** in greater detail. FIGS. **6**A-**6**D show only portions of the needle cover **113**, needle cover lock **600** and the plunger **121**. One or more other features of the medicament delivery device **200** are hidden for improved clarity.
(132) FIG. **6**A shows the needle cover lock **600** in its initial configuration, corresponding to the medicament delivery device **200** being in its initial state shown in FIG. **3**A. The needle cover **113** is in its extended position and the plunger **121** has not yet been moved axially by the collar **119**. The needle cover lock **600** comprises a flexible arm **603** having a projection **601** arranged at a free end, which extends radially outwards. When the needle cover lock **600** is in its initial configuration, the projection **601** does not engage the engaging element **602** formed at the inner surface of the needle cover **113** and so does not limit proximal movement of the needle cover **113**. FIG. **6**A shows the arm **603** at least partially received within a recessed portion **604** of the plunger **121**. The flexible arm **603** may be biased into this position (e.g., due to resiliency of the flexible arm **603**). The engaging element **602** is located distal to the projection **601**.
(133) FIG. **6**B shows the needle cover lock **600** while the medicament delivery device **200** is in its holding state shown in FIG. **3**B, wherein the needle cover lock **600** remains in its initial configuration. The needle cover **113** has moved proximally from its extended position to its retracted position, as indicated by arrow **650**, such that the engaging element **602** is now located proximal to the projection **601**. The plunger **121** has been moved distally by the collar **119**, by a distance corresponding to a partial delivery of the medicament **115**. The arm **603** remains at least partially received with the recessed portion **604** of the plunger **121**, with the arm **603** and the projection **601** having not substantially moved position. The projection **601** does not engage the engaging element **602** formed at the inner surface of the needle cover **113** and so does not limit proximal movement of the needle cover **113**.
(134) FIG. **6**C shows the needle cover lock **600** while the medicament delivery device **200** is in its resumed state shown in FIG. **3**E, wherein the medicament delivery process is complete. The needle cover **113** has remained in its retracted position, however the medicament delivery mechanism **180**, and in particular the plunger **121**, has moved the needle cover lock **600** from its initial configuration (shown in FIG. **6**B) to its locking configuration (shown in FIG. **6**C). More specifically, the plunger **121** has continued to be moved distally due to rotation of the collar **119**, causing the arm **603** of the needle cover lock **600** to traverse the recessed portion **604** of the plunger **121** until it reaches a proximal side wall **605** of the recessed portion **604**. Once the arm **603** has reached the proximal side wall **605**, continued distal movement of the plunger **121** causes the flexible arm **603** and the projection **601** to be pushed radially outwards by engagement between a raised surface **606** of the plunger **121** and the arm **603**, wherein the raised surface **606** is located proximal to the recessed portion **604** on the plunger **121**. The needle cover lock **600** is held in its locking configuration by the raised surface **606** of the plunger **121**. The proximal side wall **605** and raised surface **606** may be arranged on the plunger **121** such that they engage to move the needle cover lock **600** into the locking configuration once delivery of the medicament is complete, or shortly before delivery of the medicament is complete.
(135) FIG. **6**D shows the needle cover lock **600** while the medicament delivery device **200** is in its locked state shown in FIG. **3**F. The needle cover **113** has moved distally from its retracted position to its extended position, causing the engaging element **602** to move distally relative to the projection **601** until it engages the projection **601**. As the engaging element **602** moves distally, a distally-facing ramped surface **609** of the engaging element **602** is brought into engagement with a proximally-facing ramped surface **607** of the projection **603**. Continued distal movement of the engaging element **602** causes the distally-facing ramped surface **609** of the engaging element **602** to slide over the proximally-facing ramped surface **607** of the projection **603**, with interaction between the ramped surfaces **609**, **607** causing the engaging element **602** to move radially outwards and/or the projection **601** to be moved radially inwards (due to flexing of the engaging element **602**, needle cover **113**, projection **601** and/or flexible arm **603**). Once a blocking surface **608** of the

projection **601** passes a blocking surface **610** of the engaging element **602**, the engaging element **602** moves radially inwards and/or the projection **601** moves radially outwards. The blocking surface **608** of the projection **601** now engages the blocking surface **610** of the engaging element **602** such that proximal movement of the needle cover **113** to uncover the needle **117** is now limited (i.e., prevented).
(136) It has been generally been described throughout this disclosure that the or each flexible arm **126** has a protrusion **127** which engages the track **128**. However, it should be noted that in some examples, the or each flexible arm **126** does not have a protrusion **127** which engages the track **128**. Instead, for example, a surface of the flexible arm(s) **126** could engage the track **128**.
(137) In any of the embodiments disclosed herein, the medicament delivery device **200** may additionally have a cap **12** which covers the distal end of the needle cover **113**. The cap **12** may be coupled to the remainder of the medicament delivery device **200** (e.g., to the body **111**) when the medicament delivery device **100** is in its initial state corresponding to FIG. **3**A. The cap **12** may be removed from the remainder of the medicament delivery device **200** by the user, prior to injection. Removal of the cap **12** may also remove a rigid needle shield (RNS) surrounding the needle **117**, wherein the RNS is removed with the cap **12** (e.g., due to engagement between the RNS and the cap **12**).
(138) A method **700** of using a medicament delivery device, which may be the medicament delivery device **200** described in relation to FIGS. **3**A-**3**F, will now be described with reference to FIG. **7**.
(139) At optional operation **710** of the method **700**, a cap **12** of the medicament delivery device is removed (if present).
(140) At operation **720**, a needle cover **113** of the medicament delivery device is moved from an extended position to a retracted position to initiate delivery of a medicament **115** by a medicament delivery mechanism **180** of the medicament delivery device (e.g., as previously described herein). The needle cover **113** may be moved from the extended position to the retracted position by pressing the needle cover **113** against an injection site.
(141) At operation **730**, the needle cover **113** of the medicament delivery device is moved from the retracted position to the extended position to pause the delivery of the medicament **115** by the medicament delivery mechanism **180** (e.g., as previously described herein). The needle cover **113** may be moved from the retracted position to the extended position by removing the needle cover **113** from the injection site. The medicament delivery mechanism **180** may have delivered only a portion of the medicament **115** held within the medicament delivery device when the delivery was paused.
(142) At operation **740**, the delivery of the medicament **115** by the medicament delivery mechanism **180** is resumed (e.g., as previously described herein). To resume the medicament delivery, the user may first actuate an actuation member **170** of the medicament delivery device to rotate a needle cover guide **143**, before moving the needle cover **113** from the extended position to the retracted position (e.g., by pressing the needle cover **113** against the injection site, or a different injection site). The medicament delivery mechanism may continue delivering the remaining medicament **115**.
(143) Optionally, operations **730** and **740** may be repeated multiple times in order to pause and resume the delivery of the medicament multiple times.
(144) Once medicament delivery is complete, at operation **750**, the needle cover **113** of the medicament delivery device is moved from the retracted position to the extended position (e.g., by removing the needle cover **113** from the injection site). As the needle cover **113** is moved from the retracted position to the extended position, a needle cover lock prevents subsequent proximal movement of the needle cover **113**.
(145) While various aspects of this disclosure have been described as suitable for pausing and resuming the delivery of a medicament such that the medicament may be delivered in a plurality of

discrete doses, it should be understood that aspects of this disclosure may additionally/alternatively be used to pause and resume a medicament delivery process prior to any medicament being dispensed. For example, in accordance with one or more aspects of this disclosure, a user may initiate movement of the medicament delivery mechanism by pushing the medicament delivery device into an injection site, but may remove the medicament delivery device from the injection site prior to actual dispensing of any medicament (e.g., before the plunger has contacted the piston). Thus, no medicament is actually injected into the subject. Such a feature may be useful where the user has inserted the needle at an inappropriate injection site, since it allows the user to move the medicament delivery device to a correct injection site before any medicament is actually dispensed.

(146) While various aspects of this disclosure describe the use of an actuation member **170** to rotate the needle cover guide **143**, it should be noted that alternative methods of rotating the needle cover guide **143** to allow for proximal movement of the needle cover **113** could be employed. For example, in some embodiments at least a portion of the needle cover guide **143** may be accessible to the user from outside the body **111** such that the user may rotate the needle cover guide **143** by directly touching and rotating the needle cover guide **143** (e.g., without the use of an actuation member **170**).

(147) The terms "drug" or "medicament" are used synonymously herein and describe a pharmaceutical formulation containing one or more active pharmaceutical ingredients or pharmaceutically acceptable salts or solvates thereof, and optionally a pharmaceutically acceptable carrier. An active pharmaceutical ingredient ("API"), in the broadest terms, is a chemical structure that has a biological effect on humans or animals. In pharmacology, a drug or medicament is used in the treatment, cure, prevention, or diagnosis of disease or used to otherwise enhance physical or mental well-being. A drug or medicament may be used for a limited duration, or on a regular basis for chronic disorders.

(148) As described below, a drug or medicament can include at least one API, or combinations thereof, in various types of formulations, for the treatment of one or more diseases. Examples of API may include small molecules having a molecular weight of 500 Da or less; polypeptides, peptides and proteins (e.g., hormones, growth factors, antibodies, antibody fragments, and enzymes); carbohydrates and polysaccharides; and nucleic acids, double or single stranded DNA (including naked and cDNA), RNA, antisense nucleic acids such as antisense DNA and RNA, small interfering RNA (siRNA), ribozymes, genes, and oligonucleotides. Nucleic acids may be incorporated into molecular delivery systems such as vectors, plasmids, or liposomes. Mixtures of one or more drugs are also contemplated.

(149) The drug or medicament may be contained in a primary package or "drug container" adapted for use with a drug delivery device. The drug container may be, e.g., a cartridge, syringe, reservoir, or other solid or flexible vessel configured to provide a suitable chamber for storage (e.g., short- or long-term storage) of one or more drugs. For example, in some instances, the chamber may be designed to store a drug for at least one day (e.g., 1 to at least 30 days). In some instances, the chamber may be designed to store a drug for about 1 month to about 2 years. Storage may occur at room temperature (e.g., about 20° C.), or refrigerated temperatures (e.g., from about −4° C. to about 4° C.). In some instances, the drug container may be or may include a dual-chamber cartridge configured to store two or more components of the pharmaceutical formulation to-be-administered (e.g., an API and a diluent, or two different drugs) separately, one in each chamber. In such instances, the two chambers of the dual-chamber cartridge may be configured to allow mixing between the two or more components prior to and/or during dispensing into the human or animal body. For example, the two chambers may be configured such that they are in fluid communication with each other (e.g., by way of a conduit between the two chambers) and allow mixing of the two components when desired by a user prior to dispensing. Alternatively, or in addition, the two chambers may be configured to allow mixing as the components are being dispensed into the

human or animal body.
(150) The drugs or medicaments contained in the drug delivery devices as described herein can be used for the treatment and/or prophylaxis of many different types of medical disorders. Examples of disorders include, e.g., diabetes mellitus or complications associated with diabetes mellitus such as diabetic retinopathy, thromboembolism disorders such as deep vein or pulmonary thromboembolism. Further examples of disorders are acute coronary syndrome (ACS), angina, myocardial infarction, cancer, macular degeneration, inflammation, hay fever, atherosclerosis and/or rheumatoid arthritis. Examples of APIs and drugs are those as described in handbooks such as Rote Liste 2014, for example, without limitation, main groups 12 (anti-diabetic drugs) or 86 (oncology drugs), and Merck Index, 15th edition.
(151) Examples of APIs for the treatment and/or prophylaxis of type 1 or type 2 diabetes mellitus or complications associated with type 1 or type 2 diabetes mellitus include an insulin, e.g., human insulin, or a human insulin analogue or derivative, a glucagon-like peptide (GLP-1), GLP-1 analogues or GLP-1 receptor agonists, or an analogue or derivative thereof, a dipeptidyl peptidase-4 (DPP4) inhibitor, or a pharmaceutically acceptable salt or solvate thereof, or any mixture thereof. As used herein, the terms “analogue” and “derivative” refers to a polypeptide which has a molecular structure which formally can be derived from the structure of a naturally occurring peptide, for example that of human insulin, by deleting and/or exchanging at least one amino acid residue occurring in the naturally occurring peptide and/or by adding at least one amino acid residue. The added and/or exchanged amino acid residue can either be codable amino acid residues or other naturally occurring residues or purely synthetic amino acid residues. Insulin analogues are also referred to as “insulin receptor ligands”. In particular, the term “derivative” refers to a polypeptide which has a molecular structure which formally can be derived from the structure of a naturally occurring peptide, for example that of human insulin, in which one or more organic substituent (e.g., a fatty acid) is bound to one or more of the amino acids. Optionally, one or more amino acids occurring in the naturally occurring peptide may have been deleted and/or replaced by other amino acids, including non-codeable amino acids, or amino acids, including non-codeable, have been added to the naturally occurring peptide.
(152) Examples of insulin analogues are Gly(A21), Arg(B31), Arg(B32) human insulin (insulin glargine); Lys(B3), Glu(B29) human insulin (insulin glulisine); Lys(B28), Pro(B29) human insulin (insulin lispro); Asp(B28) human insulin (insulin aspart); human insulin, wherein proline in position B28 is replaced by Asp, Lys, Leu, Val or Ala and wherein in position B29 Lys may be replaced by Pro; Ala(B26) human insulin; Des(B28-B30) human insulin; Des(B27) human insulin and Des(B30) human insulin.
(153) Examples of insulin derivatives are, for example, B29-N-myristoyl-des(B30) human insulin, Lys(B29) (N-tetradecanoyl)-des(B30) human insulin (insulin detemir, Levemir®); B29-N-palmitoyl-des(B30) human insulin; B29-N-myristoyl human insulin; B29-N-palmitoyl human insulin; B28-N-myristoyl LysB28ProB29 human insulin; B28-N-palmitoyl-LysB28ProB29 human insulin; B30-N-myristoyl-ThrB29LysB30 human insulin; B30-N-palmitoyl-ThrB29LysB30 human insulin; B29-N—(N-palmitoyl-gamma-glutamyl)-des(B30) human insulin, B29-N-omega-carboxypentadecanoyl-gamma-L-glutamyl-des(B30) human insulin (insulin degludec, Tresiba®); B29-N—(N-lithocholyl-gamma-glutamyl)-des(B30) human insulin; B29-N-(ω-carboxyheptadecanoyl)-des(B30) human insulin and B29-N-(ω-carboxyheptadecanoyl) human insulin.
(154) Examples of GLP-1, GLP-1 analogues and GLP-1 receptor agonists are, for example, Lixisenatide (Lyxumia®), Exenatide (Exendin-4, Byetta®, Bydureon®, a 39 amino acid peptide which is produced by the salivary glands of the Gila monster), Liraglutide (Victoza®), Semaglutide, Taspoglutide, Albiglutide (Syncria®), Dulaglutide (Trulicity®), rExendin-4, CJC-1134-PC, PB-1023, TTP-054, Langlenatide/HM-11260C (Efpeglenatide), HM-15211, CM-3, GLP-1 Eligen, ORMD-0901, NN-9423, NN-9709, NN-9924, NN-9926, NN-9927, Nodexen, Viador-

GLP-1, CVX-096, ZYOG-1, ZYD-1, GSK-2374697, DA-3091, MAR-701, MAR709, ZP-2929, ZP-3022, ZP-DI-70, TT-401 (Pegapamodtide), BHM-034. MOD-6030, CAM-2036, DA-15864, ARI-2651, ARI-2255, Tirzepatide (LY3298176), Bamadutide (SAR425899), Exenatide-XTEN and Glucagon-Xten.

(155) An example of an oligonucleotide is, for example: mipomersen sodium (Kynamro®), a cholesterol-reducing antisense therapeutic for the treatment of familial hypercholesterolemia or RG012 for the treatment of Alport syndrom.

(156) Examples of DPP4 inhibitors are Linagliptin, Vildagliptin, Sitagliptin, Denagliptin, Saxagliptin, Berberine.

(157) Examples of hormones include hypophysis hormones or hypothalamus hormones or regulatory active peptides and their antagonists, such as Gonadotropine (Follitropin, Lutropin, Choriongonadotropin, Menotropin), Somatropine (Somatropin), Desmopressin, Terlipressin, Gonadorelin, Triptorelin, Leuprorelin, Buserelin, Nafarelin, and Goserelin.

(158) Examples of polysaccharides include a glucosaminoglycane, a hyaluronic acid, a heparin, a low molecular weight heparin or an ultra-low molecular weight heparin or a derivative thereof, or a sulphated polysaccharide, e.g. a poly-sulphated form of the above-mentioned polysaccharides, and/or a pharmaceutically acceptable salt thereof. An example of a pharmaceutically acceptable salt of a poly-sulphated low molecular weight heparin is enoxaparin sodium. An example of a hyaluronic acid derivative is Hylan G-F 20 (Synvisc®), a sodium hyaluronate.

(159) The term “antibody”, as used herein, refers to an immunoglobulin molecule or an antigen-binding portion thereof. Examples of antigen-binding portions of immunoglobulin molecules include F(ab) and F(ab′)2 fragments, which retain the ability to bind antigen. The antibody can be polyclonal, monoclonal, recombinant, chimeric, de-immunized or humanized, fully human, non-human, (e.g., murine), or single chain antibody. In some embodiments, the antibody has effector function and can fix complement. In some embodiments, the antibody has reduced or no ability to bind an Fc receptor. For example, the antibody can be an isotype or subtype, an antibody fragment or mutant, which does not support binding to an Fc receptor, e.g., it has a mutagenized or deleted Fc receptor binding region. The term antibody also includes an antigen-binding molecule based on tetravalent bispecific tandem immunoglobulins (TBTI) and/or a dual variable region antibody-like binding protein having cross-over binding region orientation (CODV).

(160) The terms “fragment” or “antibody fragment” refer to a polypeptide derived from an antibody polypeptide molecule (e.g., an antibody heavy and/or light chain polypeptide) that does not comprise a full-length antibody polypeptide, but that still comprises at least a portion of a full-length antibody polypeptide that is capable of binding to an antigen. Antibody fragments can comprise a cleaved portion of a full length antibody polypeptide, although the term is not limited to such cleaved fragments. Antibody fragments that are useful in the present disclosure include, for example, Fab fragments, F(ab′)2 fragments, scFv (single-chain Fv) fragments, linear antibodies, monospecific or multispecific antibody fragments such as bispecific, trispecific, tetraspecific and multispecific antibodies (e.g., diabodies, triabodies, tetrabodies), monovalent or multivalent antibody fragments such as bivalent, trivalent, tetravalent and multivalent antibodies, minibodies, chelating recombinant antibodies, tribodies or bibodies, intrabodies, nanobodies, small modular immunopharmaceuticals (SMIP), binding-domain immunoglobulin fusion proteins, camelized antibodies, and VHH containing antibodies. Additional examples of antigen-binding antibody fragments are known in the art.

(161) The terms “Complementarity-determining region” or “CDR” refer to short polypeptide sequences within the variable region of both heavy and light chain polypeptides that are primarily responsible for mediating specific antigen recognition. The term “framework region” refers to amino acid sequences within the variable region of both heavy and light chain polypeptides that are not CDR sequences, and are primarily responsible for maintaining correct positioning of the CDR sequences to permit antigen binding. Although the framework regions themselves typically do not

directly participate in antigen binding, as is known in the art, certain residues within the framework regions of certain antibodies can directly participate in antigen binding or can affect the ability of one or more amino acids in CDRs to interact with antigen.
(162) Examples of antibodies are anti PCSK-9 mAb (e.g., Alirocumab), anti IL-6 mAb (e.g., Sarilumab), and anti IL-4 mAb (e.g., Dupilumab).
(163) Pharmaceutically acceptable salts of any API described herein are also contemplated for use in a drug or medicament in a drug delivery device. Pharmaceutically acceptable salts are for example acid addition salts and basic salts.
(164) Those of skill in the art will understand that modifications (additions and/or removals) of various components of the APIs, formulations, apparatuses, methods, systems and embodiments described herein may be made without departing from the full scope and spirit of the present disclosure, which encompass such modifications and any and all equivalents thereof.
(165) An example drug delivery device may involve a needle-based injection system as described in Table 1 of section 5.2 of ISO 11608-1:2014(E). As described in ISO 11608-1:2014(E), needle-based injection systems may be broadly distinguished into multi-dose container systems and single-dose (with partial or full evacuation) container systems. The container may be a replaceable container or an integrated non-replaceable container.
(166) As further described in ISO 11608-1:2014(E), a multi-dose container system may involve a needle-based injection device with a replaceable container. In such a system, each container holds multiple doses, the size of which may be fixed or variable (pre-set by the user). Another multi-dose container system may involve a needle-based injection device with an integrated non-replaceable container. In such a system, each container holds multiple doses, the size of which may be fixed or variable (pre-set by the user).
(167) As further described in ISO 11608-1:2014(E), a single-dose container system may involve a needle-based injection device with a replaceable container. In one example for such a system, each container holds a single dose, whereby the entire deliverable volume is expelled (full evacuation). In a further example, each container holds a single dose, whereby a portion of the deliverable volume is expelled (partial evacuation). As also described in ISO 11608-1:2014(E), a single-dose container system may involve a needle-based injection device with an integrated non-replaceable container. In one example for such a system, each container holds a single dose, whereby the entire deliverable volume is expelled (full evacuation). In a further example, each container holds a single dose, whereby a portion of the deliverable volume is expelled (partial evacuation).
(168) Those of skill in the art will understand that modifications (additions and/or removals) of various components of the embodiments described herein may be made without departing from the full scope and spirit of the present disclosure, which encompass such modifications and any and all equivalents thereof.
LIST OF REFERENCE NUMBERS
(169) **10**—drug delivery device **11**—housing **11***a*—window **12**—cap assembly **13**—needle sleeve **14**—reservoir **15**—medicament **17**—needle **20**—distal region **21**—proximal region **22**—button **23**—bung or piston **100**—medicament delivery device **111**—body **113**—needle cover **114**—container **115**—medicament **117**—needle **118**—needle cover biasing member **119**—collar **120**—distal end (of the needle cover) **121**—plunger **122**—external screw thread **123**—bung or piston **124**—drive member **126**—arm **127**—protrusion (of needle cover guide) **128**—track **130**—proximal end (of body) **131**—distal end (of body) **140**—distal end (of needle) **142**—arrow **143**—needle cover guide **144**—axis **150**—syringe **160**—latch **161**—engaging element (of latch) **162**—flexible arm (of latch) **166**—side wall (of engaging element of latch) **170**—actuation member **171**—spring **180**—medicament delivery mechanism **191**—engaging element (of collar) **196**—side wall (of engaging element of collar) **200**—medicament delivery device **301**—needle cover recess **302**—ramped side wall **501**—first position (of track) **502**—second position (of track) **503**—third position (of track) **504**—fourth position (of track) **511**—first region (of track) **512**—second region (of track) **513**—

third region (of track) **514**—fourth region (of track) **520**—arrow **521**—arrow **530**—proximal side wall (of first region) **540**—locking element (of track) **541**—ramped surface (of locking element) **542**—locking surface (of locking element) **543**—side wall (of third region) **560***a*-**560***d*—portions (of track) **580***a*-**580***d*—engagement elements (of actuation member) **581***a*-**581***d*—engagement elements (of needle cover guide) **582***a*-**582***d*—ramped surfaces **583***a*-**583***d*—ramped surfaces **600**—needle cover lock **601**—projection (of needle cover lock) **602**—engaging element (of needle cover) **603**—flexible arm (of needle cover lock) **604**—recessed portion (in plunger) **605**—proximal side wall (of recessed portion) **606**—raised surface (of plunger) **607**—ramped surface (of needle cover lock) **608**—blocking surface (of needle cover lock) **609**—ramped surface (of engaging element of needle cover) **610**—blocking surface (of engaging element of needle cover) **650**—arrow **700**—method **710**—operation **720**—operation **730**—operation **740**—operation **750**—operation

Claims

1. A medicament delivery device comprising: a needle for injecting a medicament; a body having a proximal end and a distal end; a needle cover axially movable relative to the body between an extended position, in which a distal end of the needle cover is distal to a distal end of the needle, and a retracted position, in which the distal end of the needle is distal to the distal end of the needle cover; a needle cover biasing member configured to bias the needle cover distally; a medicament delivery mechanism; and a latch movable by the needle cover between an engaged configuration, in which the latch is engaged with a component of a medicament delivery mechanism to prevent delivery of the medicament, and a disengaged configuration, in which the latch is disengaged from the component of the medicament delivery mechanism to allow delivery of the medicament, wherein the needle cover is configured such that: proximal movement of the needle cover from the extended position towards the retracted position moves the latch from the engaged configuration to the disengaged configuration; and distal movement of the needle cover from the retracted position towards the extended position moves the latch from the disengaged configuration to the engaged configuration, and wherein the latch comprises an engaging element configured to: engage an engaging element of the component of the medicament delivery mechanism when the latch is in the engaged position; and be disengaged from the engaging element of the component when the latch is in the disengaged position.
2. The medicament delivery device according to claim 1, wherein the latch is movable between the engaged position and the disengaged position a plurality of times for pausing and resuming delivery of the medicament a plurality of times.
3. The medicament delivery device according to claim 1, further comprising a needle cover guide comprising a track, wherein the needle cover comprises an arm configured to engage the track such that a proximal movement of the needle cover is limited after the needle cover has moved from the retracted position to the extended position.
4. The medicament delivery device according to claim 3, wherein the needle cover guide is rotatable relative to the body, and wherein the track extends at least partially around a circumference of the needle cover guide for rotating the needle cover guide as the needle cover moves from the extended position to the retracted position.
5. The medicament delivery device according to claim 3, wherein the arm comprises a protrusion configured to engage the track.
6. The medicament delivery device according to claim 5, wherein the track comprises a first region and a second region, wherein the protrusion travels from the first region to the second region as the needle cover moves from the extended position to the retracted position.
7. The medicament delivery device according to claim 6, wherein: the track comprises a third region, the protrusion travels from the second region to the third region as the needle cover moves distally from the retracted position to the extended position, and the third region comprises a

locking element configured to limit proximal movement of the needle cover when the needle cover is in the extended position.

8. The medicament delivery device according to claim 7, wherein the locking element comprises a locking surface configured to be engaged by the needle cover when the needle cover is in the extended position to limit proximal movement of the needle cover.

9. The medicament delivery device according to claim 7, wherein the track further comprises a fourth region, wherein the protrusion travels from the third region to the fourth region as the needle cover guide is rotated, wherein proximal movement of the needle cover from the locked position to the retracted position is allowed when the protrusion is in the fourth region.

10. The medicament delivery device according to claim 9, further comprising an actuation member, wherein the actuation member is actuatable by a user to rotate the needle cover guide relative to the needle cover such that the protrusion travels from the third region to the fourth region.

11. The medicament delivery device of claim 10, wherein the actuation member comprises a button arranged at a proximal end of the medicament delivery device.

12. The medicament delivery device of claim 1, wherein the engaging element of the latch comprises a projection and the engaging element of the component comprises a recess.

13. The medicament delivery device according to claim 1, wherein the component of the medicament delivery mechanism is a collar that is rotatable relative to the body to dispense the medicament, wherein the latch is configured to engage the collar when in the engaged position to limit rotation of the collar.

14. The medicament delivery device according to claim 1, wherein the latch is biased from the engaged configuration to the disengaged configuration, wherein the latch is held in the engaged position by the needle cover when the needle cover is in the extended position, and wherein the needle cover comprises a recess configured to receive at least a portion of the latch when the needle cover is in the retracted position such that the latch can move to the disengaged position.

15. The medicament delivery device according to claim 1, further comprising a needle cover lock that is movable by the medicament delivery mechanism between an initial configuration, in which movement of the needle cover from the extended position to the retracted position is not limited by the needle cover lock, and a locking position, in which movement of the needle cover from the extended position to the retracted position is limited by the needle cover lock.

16. The medicament delivery device of claim 15, wherein the needle cover lock is moved from the disengaged position to the engaged position by a plunger of the medicament delivery mechanism.

17. The medicament delivery device according to claim 1, further comprising the medicament.

18. A method of operating a medicament delivery device, the method comprising: moving a needle cover of the medicament delivery device from an extended position to a retracted position to initiate delivery of a medicament; moving the needle cover of the medicament delivery device from the retracted position to the extended position to pause the delivery of the medicament; and resuming the delivery of the medicament by: actuating an actuation member of the medicament delivery device to rotate a needle cover guide, wherein at least one of the actuation member or the needle cover guide comprises a ramped surface configured to convert linear movement of the actuation member into rotation of the needle cover guide; and subsequent to actuating the actuation member, moving the needle cover from the extended position to the retracted position.